# 综合练习

## 人员

王承周、王毅博、许睿谦、阮文璋 到课

## 上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/3158

比赛概况 题目列表 选择题列表 提交记录 ★ 实时榜单 ★ 选择题排行榜 编辑比赛

**王向东老师周日一点半C++综合练习**

刷新

| # | 用户名 | 姓名 | 编程分 | 时间 | A | B | C |
|---|---|---|---|---|---|---|---|
| 1 | ruanwenzhang | 阮文璋 | 300 | 7573 | 100 | 100 | 100 |
| 2 | wangyibo | 王毅博 | 300 | 20623 | 100 | 100 | 100 |
| 3 | wangchengzhou | 王承周 | 300 | 22566 | 100 | 100 | 100 |
| 4 | yangjunyan | 杨俊彦 | 300 | 28864 | 100 | 100 | 100 |
| 5 | xuruiqian | 许睿谦 | 200 | 5073 | 100 | 100 | |
| 6 | chujinxuan | 褚锦轩 | 200 | 12835 | 100 | 100 | |

## 本周作业

https://cppoj.kids123code.com/contest/3195 (课上讲了 A ~ C 题, 课后作业是 D 题)

## 课堂表现

今天课上的 3 道题目都不是很简单, 需要一些思维, 建议同学们课后好好复盘一下这几道题。

## 课堂内容

### Subsequence Path

dp, 按题目输入的 K 条边的顺序一条边一条边的看

```
#include <bits/stdc++.h>
#define int long long

using namespace std;

const int maxn = 2e5 + 5;
const int inf = 0x3f3f3f3f3f3f3f3f;
struct node {
  int from, to, value;
} e[maxn];
int f[maxn];

signed main()
```

```
{
  int n, m, K; cin >> n >> m >> K;
  for (int i = 1; i <= m; ++i) cin >> e[i].from >> e[i].to >> e[i].value;

  memset(f, 0x3f, sizeof(f));
  f[1] = 0;
  for (int i = 1; i <= K; ++i) {
    int id; cin >> id;
    int from = e[id].from, to = e[id].to, value = e[id].value;
    f[to] = min(f[to], f[from]+value);
  }

  cout << (f[n]==inf ? -1 : f[n]) << endl;
  return 0;
}
```

**graph**

二分 + bfs搜索

```
#include <bits/stdc++.h>
#define int long long

using namespace std;

const int maxn = 2e5 + 5;
vector<int> vec[maxn];
int w[maxn], c[maxn], c2[maxn];
bool st[maxn];
int n, m;

bool check(int mid) {
    for (int i = 1; i <= n; ++i) c2[i] = c[i];
    memset(st, false, sizeof(st));

    queue<int> q;
    for (int i = 1; i <= n; ++i) {
        if (c2[i] <= mid) q.push(i), st[i] = true;
    }
    while (!q.empty()) {
        int u = q.front(); q.pop();
        for (int i : vec[u]) {
            c2[i] -= w[u];
            if (!st[i] && c2[i]<=mid) q.push(i), st[i] = true;
        }
    }

    for (int i = 1; i <= n; ++i) {
        if (!st[i]) return false;
    }
    return true;
```

```
}

signed main()
{
    cin >> n >> m;
    for (int i = 1; i <= n; ++i) cin >> w[i];
    while (m -- ) {
        int a, b; cin >> a >> b;
        vec[a].push_back(b), vec[b].push_back(a);
        c[a] += w[b], c[b] += w[a];
    }

    int l = 0, r = 2e14;
    while (l <= r) {
        int mid = (l + r) / 2;
        if (check(mid)) r = mid-1;
        else l = mid+1;
    }
    cout << l << endl;
    return 0;
}
```

**Lucky Numbers**

枚举后面的 ai 依次为幸运数字时, 什么时候在 a1 处能重合最多

```
#include <bits/stdc++.h>
#define int long long

using namespace std;

const int maxn = 1e5 + 5;
int w[maxn], c[maxn];
int val[15];

signed main()
{
  int n, m; cin >> n >> m;
  for (int i = 1; i <= n-1; ++i) {
    cin >> w[i];
    if (i&1) c[i] = c[i-1] + w[i];
    else c[i] = c[i-1] - w[i];
  }
  for (int i = 1; i <= m; ++i) cin >> val[i];

  map<int, int> mp;
  for (int i = 2; i <= n; ++i) {
    for (int j = 1; j <= m; ++j) {
      int x;
      if (i&1) x = c[i-1] + val[j];
      else x = c[i-1] - val[j];
```

```
        mp[x]++;
      }
    }

    int maxx = 0;
    for (auto it : mp) {
      int a = it.first, b = it.second, cnt = 0;
      for (int j = 1; j <= m; ++j) {
        if (val[j] == a) cnt = 1;
      }
      maxx = max(maxx, b+cnt);
    }
    cout << maxx << endl;
    return 0;
  }
```